

MEGAMINX

12 SIDES · 50 PIECES · ONE SECRET

SOLUTION GUIDE



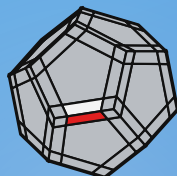
UNLOCK THE SECRET!

DEFINITIONS OF MEGAMINX PIECES

THE PARTS:

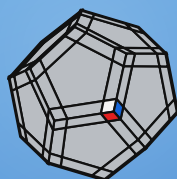
EDGE PIECES

PIECES WITH TWO (2) COLORS. THERE ARE THIRTY (30) EDGE PIECES. FIVE OF THEM HAVE A WHITE SIDE — THEY MAKE THE WHITE STAR.



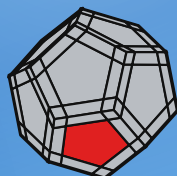
CORNER PIECES

PIECES WITH THREE (3) COLORS. THERE ARE TWENTY (20) CORNER PIECES.



CENTER PIECES

PIECES WITH ONE (1) COLOR. THERE ARE TWELVE (12), ONE ON EACH FACE. THEY **DO NOT MOVE** AND REPRESENT THE COLOR OF THEIR FACE.



A megaminx has **12 faces** and **12 colors**. Each color appears on exactly one face. **WHITE** and **GRAY** are on opposite faces — you will start with the white face and finish with the gray one. (Some megaminxes use slightly different colors — use the face colors of YOUR puzzle; the steps are the same.)

EACH FACE OF THE MEGAMINX IS REPRESENTED BY A LETTER

- U** = **Up Face** — the top face
- F** = **Front Face** — the face facing you, below the top
- R** = **Right Face** — the face just right of the front
- L** = **Left Face** — the face just left of the front
- D** = **Down Face** — the face below and between F and R

A letter with an "i" after it means an **inverted** (counter-clockwise) turn of that face. A letter alone means a clockwise turn (as you look at that face). Every turn is **one click** — one fifth of the way around.

VERY IMPORTANT

The pictures show exactly which face moves and which way. Hold your megaminx to match the picture before each sequence. Dark gray on a picture means **the color there does not matter**.



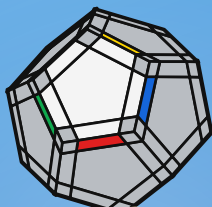
STAGE 2:

SOLVE THE WHITE STAR

HOLDING YOUR MEGAMINX: Hold your megaminx with the **WHITE** center piece on top (**U**). You will make a white star, and the side color of each star edge must match its side center piece.

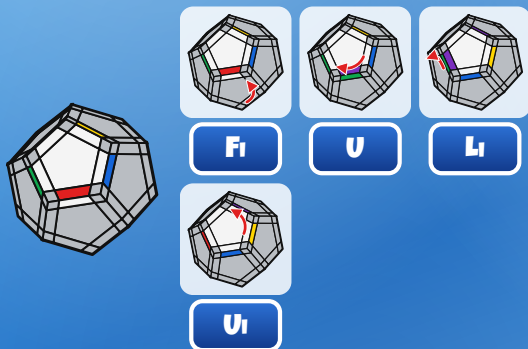
TIPS:

- Find a white edge in the bottom part of the puzzle. Turning the bottom faces never disturbs finished star edges on top.
- Park it on the **bottom-right edge of the matching side face**, white sticker facing **down-right**; then turn that side face **twice counter-clockwise** — the edge rises into the star.
- Parked on the **bottom-left** with white facing down-left? Turn the side face **twice clockwise** instead. This stage is practice and trial-and-error, just like the white cross on your cube!



YOUR GOAL

If an edge is **in the star but Flipped** (white facing out, like the picture), hold the puzzle so that edge is at the front and do this sequence:



CONGRATULATIONS!

If the top of your megaminx shows a white star with every side color matching its center, you can move to **Stage 3!**

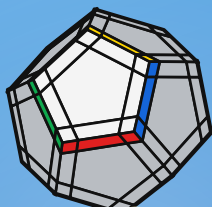
STAGE 3:

SOLVE THE WHITE CORNERS

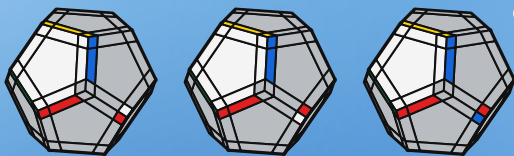
HOLDING YOUR MEGAMINX: Keep the **WHITE** star on top (**U**). Now you will fill in the five white corners, one at a time.

TIPS:

- Find a white corner in the bottom part of the puzzle and turn the bottom faces until it sits **directly below its place** (between the two side colors that match it).
- Hold the puzzle so that corner is at the **front-right**, just below the top, like the three pictures below. The white sticker can face three ways — the sequence works for all of them!



YOUR GOAL



Now do this sequence **over and over** (2, 4 or up to 6 times) **UNTIL** the corner pops into place with white on top:



Ri



Di



R



D

NOTE! If a white corner is stuck in the top layer in the wrong spot or twisted, hold the puzzle so it is at the top-front-right and do the sequence **once** — it will pop out below. Then place it the normal way.

CONGRATULATIONS!

If the whole white face and its side colors are solid, you are ready for **Stage 4!**

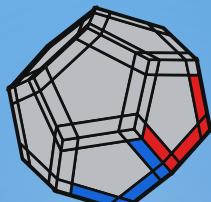
STAGE 4:

SOLVE THE SECOND ROW OF EDGES

HOLDING YOUR MEGAMINX: Now **tilt the puzzle!** Hold it so the empty edge slot is at the **front** and the face just **below** the slot points **up** — that is your new top (**U**):

TIPS:

- Turn the new top (**U**) until the edge sits above the front face with its **front color matching** the front center — a straight line, just like on your cube!
- The edge's top color points where it must go: toward (**R**) → sequence 1; toward (**L**) → sequence 2.



1) EDGE GOES DOWN-AND-RIGHT:



U



R



U_l



R_l



U_l



F_l



U

HOLD IT LIKE THIS



F

2) EDGE GOES DOWN-AND-LEFT:



U_l



L_l



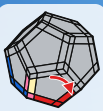
U



L



U



F



U_l



F_l

CONGRATULATIONS!

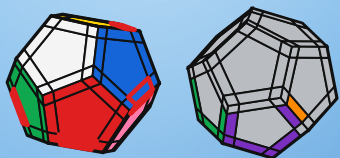
Do all five edges of the second row, then on to **Stage 5!**

STAGE 5:

THE MIDDLE BAND

1ST STEP: PLACE THE FIVE LEANING EDGES

Around the middle of the puzzle there are slanted edge slots. In this step you place **one slanted edge above each petal** — always the one that **leans the same way**, shown outlined in red:



TIPS:

- Tilt the puzzle so the empty slanted slot is at the front (second picture) and use exactly the same sequences as Stage 4.
- Go all the way around the puzzle: five slanted edges, one per petal. **Only these five!** The other five slanted slots get filled in the 2nd Step together with the corners.



F

Then continue straight to the 2nd Step on the next page.

STAGE 5:

THE MIDDLE BAND

2nd STEP: CORNER + EDGE PAIRS

HOLDING YOUR MEGAMINX: Hold the puzzle so the empty petal pocket (a corner slot plus its last slanted edge slot) is at the **top front-right**. Just like Stage 3 — but the corner brings its edge along for the ride!

TIPS:

- Park the corner **below its slot**, and its edge in one of the two parking spots shown — pick by which way the corner's **top color** faces:



repeat sequence 1x



repeat sequence 5x

Then repeat the Stage-3 sequence **until corner and edge BOTH snap in together** (up to 6 times):



R_I



D_I



R



D

NOTE! Top color facing **down**? Do the sequence **once** and look again — it will match a picture. Piece stuck in a wrong slot? Pop it out as in Stage 3.

CONGRATULATIONS!

Five pockets done? The middle band is finished — on to Stage 6!

STAGE 6:

THE LAST ROW OF CORNERS AND EDGES

1ST STEP: THE FIVE LOW CORNERS

Tilt the puzzle even further: hold it so an empty low corner slot is at the top front-right (picture). Park the matching corner below it and repeat the Stage-3 sequence until it pops in — **exactly like before**. The staging area is now next to the **GRAY** face, which is still unsolved, so nothing can break.



R_i



D_i



R



D

2ND STEP: THE FIVE RIDGE EDGES

Now turn the puzzle so **GRAY** is **on top** — and keep it there until the very end! The last five non-gray edges sit just below the gray face. Use the Stage-4 sequences: turn the gray top until the edge makes its straight line with the front center, then drop it right or left:



U



R



U_i



R_i



U_i



F_i



U



F

CONGRATULATIONS!

Everything except the gray top layer is done — you are two thirds of the way! On to **Stage 7**.

STAGE 7:

MAKE THE GRAY STAR

HOLDING YOUR MEGAMINX: **GRAY** stays on top (**U**). Look only at the five **gray edge stickers** on top: how many face up?

TIPS:

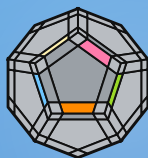
- You will always see exactly **1, 3 or 5** gray stickers facing up. (5 means the star is done — skip to Stage 8!)
- Match your puzzle to one of the pictures, hold it exactly that way, and do the sequence below. Re-count and repeat until the star is complete. The side colors do NOT need to match yet!



1 gray edge: hold it at the **back-right**; sequence once, then re-match.



3 gray; the 2 odd ones **side by side**: hold them at **front + right**; sequence once → star!



3 gray; the 2 odd ones **apart**: hold them at **front + back-right**; sequence **twice** → star!



F



U



R



U₁



R₁



F₁

CONGRATULATIONS!

Gray star on top? Excellent — Stage 8!

STAGE 8:

PUT THE STAR EDGES IN THEIR PLACES

HOLDING YOUR MEGAMINX: **GRAY** on top. Now make every star edge's side color match its center.

TIPS:

- Spin the gray top (**U**) until **as many edges as possible** match their side centers.
- Hold the puzzle so the two matching edges are at the **front** and **front-left** — the three wrong ones to the right and back, like the picture. Then do the sequence; it swaps the three wrong edges around. You may need it twice.
- If you can't find a hold with two matching edges, do the sequence once from any position and look again.



CONGRATULATIONS!

Star perfect all around? **Stage 9** awaits!

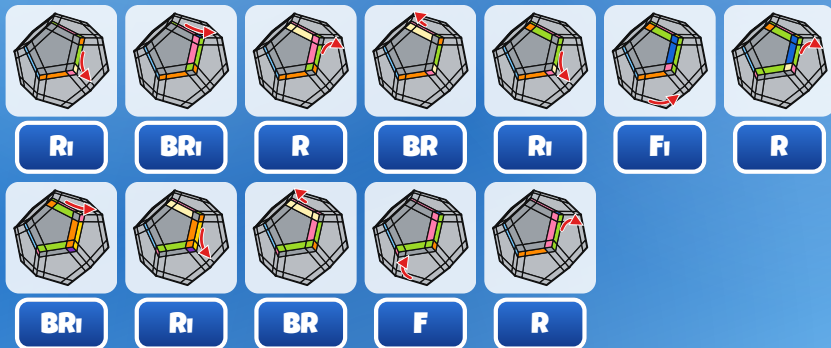
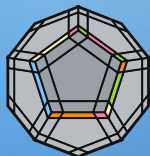
STAGE 9:

PUT THE GRAY CORNERS IN THEIR PLACES

HOLDING YOUR MEGAMINX: **GRAY** on top. Now move each gray corner **between** the two side colors it matches. Twisted is OK — location is all that counts! The sequence below cycles the **front-right, right and back corners**; the two **left corners never move**. Count how many corners sit between their right colors, then:

TIPS:

- **2 correct, side by side:** hold them as the two LEFT corners. Sequence once — not done? once more.
- **2 correct, apart:** hold one just right of the back corner and one at back-left. Sequence once, then re-count.
- **1 correct:** hold it at the FRONT-LEFT. Sequence once, re-count.
- **0 correct:** sequence once from anywhere; if nothing improved, same hold once more. Re-count.



CONGRATULATIONS!

All five corners between their right colors? One stage to go!

STAGE 10:

TWIST THE GRAY CORNERS

HOLDING YOUR MEGAMINX: **GRAY** on top. Hold the puzzle with a twisted corner at the **top front-right** and repeat the good old Stage-3 sequence until its gray sticker faces up (2 or 4 times):

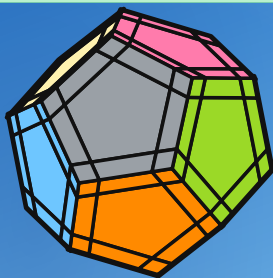


TIPS:

- Then turn **ONLY the gray top (U)** to bring the next twisted corner to the front-right, and repeat.
- **DON'T PANIC!** The rest of the puzzle will look scrambled while you work — it fixes itself the moment the last corner is twisted right. Trust the moves, just like on your cube!
- When all corners are done, spin the gray top to line everything up.

CONGRATULATIONS!

YOU DID IT! Your megaminx is solved — all 12 colors, all 50 pieces. You have unlocked the secret!



NOW YOU KNOW THE SECRET...

FUN FACTS!

The megaminx was invented around 1982, soon after the Rubik's Cube craze began. Early versions were sold under names like the **Hungarian Supernova**.

A megaminx has about **100,669,616,553,523,347,122,516,032,313,645,505,168,688,116,411,019,768,627,200,000,000,000** possible positions — that's a 1 followed by 68 digits. The Rubik's Cube has "only" 43 quintillion!

The world record for solving a megaminx is under **25 seconds**. Every solve in this booklet was checked, move by move, on a computer model of the puzzle — so if you follow the pictures exactly, it always works.

The method you just learned is the same idea as your cube's: solve a star instead of a cross, drop in corners with the Righty moves, pair up the middle, then finish the last layer in four small steps. Bigger puzzle — same secret.



Made for a young puzzler who already unlocked the cube. Happy twisting!